

Clinical Risk Assessment Report: Patient 17

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 10.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.49x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 1.5%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 77.0 [Top 6%] (-1.5%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.69 [Top 3%] (-1.3%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: -0.6 [Bottom 23%] (-1.2%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Minimum Metabolic Activity (SUVmin) (CONVENTIONAL_SUVbwmin) **Value: -1.01 [Bottom 9%] (-1.1%)**

Indicates the areas of lowest metabolic activity within the tumor, which may represent necrotic or poorly perfused regions.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.33 [Top 3%] (-0.7%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -1.3 [Bottom 8%] (-0.5%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -1.23 [Bottom 10%] (-0.4%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: -2.79 [Bottom <1%] (-0.3%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)

